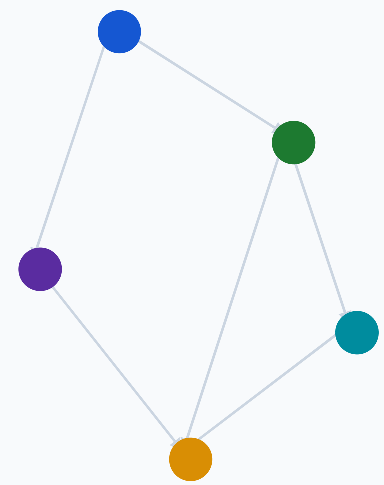
DASC690 - Capstone Project

MedGraphRAG

A traceable pipeline from text to graph evidence to cited answers

Raw literature → structured graph evidence → evidence-grounded QA



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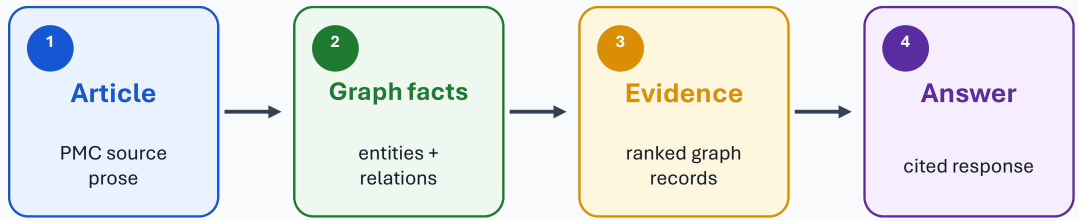
Research Question

Can biomedical prose become graph evidence for traceable question answering?

Focus

- Domain: PMC literature on lipids and cardiovascular disease.
- Output: cited answer or abstention when evidence is insufficient.
- Core contribution: keep answers linked to source evidence.

Article → Graph Facts → Evidence → Answer



The key test is not whether the system can answer, but whether each answer can be traced back to graph evidence and source provenance.

Why Transform the Text?

Biomedical prose must become typed records before the system can retrieve evidence by entity and relationship.

What source text contains

- Findings can span long passages.
- Aliases can name one concept:
  - LDL-C and low-density lipoprotein cholesterol.
  - TG and triglycerides
- Relations can include direction, negation, and qualifiers.
- The pipeline must keep PMCID and chunk provenance.

Actual processed evidence

More potent agents (eg, atorvastatin, rosuvastatin) were later developed ... up to 55% reduction in LDL.

Extracted edge: atorvastatin - REDUCES -> LDL cholesterol.

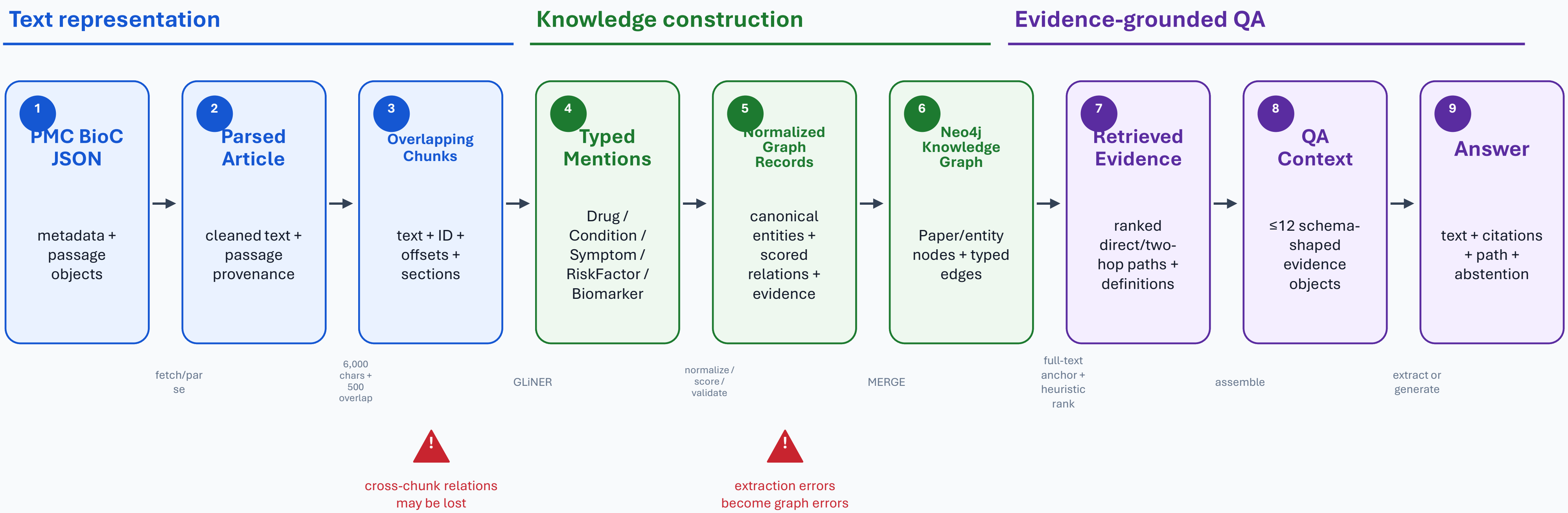
Source: PMC2952453, chunk 0002.

Drug **Biomarker** **Relation cue** **PMCID + chunk**

The pipeline selects typed mentions, scores a REDUCES relation, and keeps the source location.

End-to-End Data Transformation

One pipeline converts article text into graph records, retrieved evidence, and cited answers.



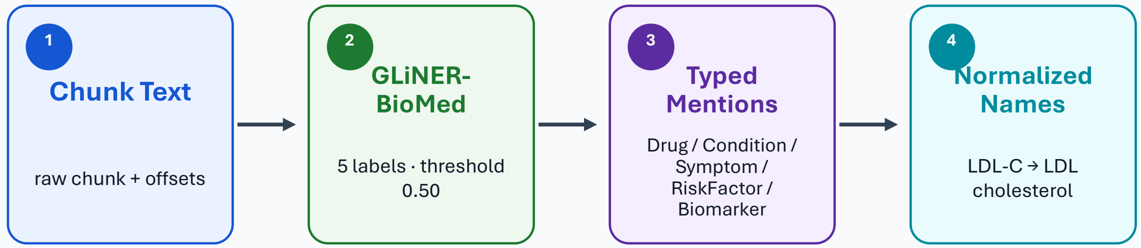
Pipeline rule: keep each transformation inspectable and tied to source evidence.

Entity Extraction From Text

GLiNER-BioMed identifies typed biomedical mentions; normalization happens after detection.

How it works

- Text is split into chunks. GLiNER-BioMed reads each chunk.
- Model: Ihor/gliner-biomed-small-v1.0.
- Allowed labels: Drug, Condition, Symptom, RiskFactor, Biomarker.
- Default entity threshold: 0.50.
- Mention spans keep type, text, offsets, and confidence during scoring.
- Normalized records keep the mention, canonical name, confidence, method, and score.
- Normalization uses an exact alias, MiniLM similarity, or a cleaned surface form.



Effective call

```
predict_entities(
  chunk.text,
  ["Drug", "Condition",
"Symptom",
   "RiskFactor", "Biomarker"],
  threshold=0.50
)
```

What "prompted" means here

- GLiNER is label-conditioned.
- It is not instruction-prompted.
- The chunk text provides evidence.
- The threshold filters weak mentions.

Relationship Extraction & Graph Loading

Validated entity pairs become evidence-bearing graph edges in Neo4j.

1. **Normalized Mentions** — type + text + offsets + confidence
2. **Candidate Pairs** — same or adjacent sentence; ≤300 chars
3. **MiniLM Scorer** — prototypes + cues + proximity
4. **Ontology Validator** — allowed relation, endpoint types, direction, evidence
5. **Accepted Edge** — best result per relation type and endpoint pair
6. **Neo4j MERGE** — idempotent Paper, entity, and edge writes

Relationship scorer

Score =

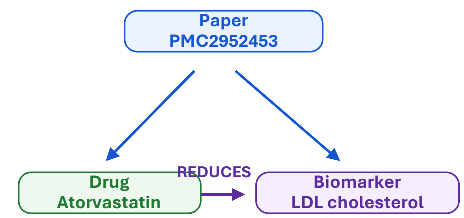
0.50	0.25	0.10	0.15
semantic	cue	proximity	entity conf.

Default threshold: 0.66; selected type overrides: 0.58-0.60

Graph build rule

- Papers become Paper nodes.
- Biomedical entities become typed nodes.
- MENTIONS links Paper -> entity.
- Typed relationships become evidence edges.
- Repeated loads update matching IDs.

Native graph shape example



Evidence fields: PMCID - chunk ID - sentence - confidence

Graph Answer Layer

Question terms are anchored to Neo4j graph nodes, converted into evidence objects, and formatted into answer JSON.

1. **Question** — user question
2. **Terms + aliases** — normalize, expand, remove generic terms
3. **Graph anchors** — full-text index, then substring fallback
4. **Graph evidence** — direct edges; optional two-hop paths
5. **QA Context** — up to 12 structured evidence records
6. **Answer JSON** — answer, sources, path, confidence, abstention

GraphRetriever

- Expands question terms with terminology aliases and matched definition terms.
- Searches biomedical_name_alias_fulltext_idx.
- Falls back to name substring search.
- Excludes MENTIONS edges from answer evidence.

Evidence object

- source entity → relationship → target entity.
- Evidence text, confidence, PMCID, chunk ID, title.
- pathId, pathStep, pathLength for two-hop paths.
- Evidence kind is graph or definition.

GraphRAGAnswerer

- Deduplicates evidence and abstains when none is found.
- Uses extractive answers for some graph and definition questions.
- The default app profile uses qwen2.5:7b-instruct for generated answers.
- The prompt permits only retrieved evidence.

Output schema: answer · sources · reasoningPath · confidence · abstained

Two Evaluation Layers

Extraction quality and QA quality are measured separately so bottlenecks stay visible.

Extraction evaluation

entity + relation precision / recall / F1

0.528
Entity F1

0.086
Relation F1
bottleneck

Text → Ent → Rel → Graph

Relation extraction is the identified weak step.

QA evaluation

retrieval, fact coverage, citations, paths, abstention

0.333
Retrieval recall
six-question dev run

0.333
Answer accuracy
six-question dev run

Q → Ret → Ctx → Ans

Reproducibility layer: DVC pins experiment state; MLflow compares parameters, metrics, and artifacts.

Key Decisions and Trade-offs

Each transformation adds structure and can remove useful detail.

Use the metrics to separate design trade-offs from model failures.

Decision	Intended benefit	Cost / Risk
1 **6,000 / 500 chunks**	Keeps more nearby context in each chunk.	Introduces pair ambiguity; cross-chunk relationships may be lost.
2 **Strict ontology**	Supports fixed validation and predictable graph queries.	Loses nuance and drops useful out-of-schema facts.
3 **Local non-instruction extraction**	Runs locally and exposes each score component.	Current weak step: relation extraction has low F1.
4 **Name normalization**	Merges known aliases into one normalized name.	False merges and false splits can create graph errors.
5 **Graph paths**	Returns ordered graph evidence for reasoning-path display.	QA is limited by incomplete graph coverage.

Current bottleneck: relationship extraction - Extraction F1: entity 0.528, relation 0.086 - QA dev: retrieval 0.333, answer accuracy 0.333

Conclusion and Next Experiment

Raw PMC prose → traceable graph evidence → cited QA.

1. **PMC prose** — raw article text →
2. **Entity extraction** — typed mentions →
3. **Relation extraction** — semantics + direction →
4. **Graph evidence** — Neo4j edges →
5. **Retrieved context** — ranked evidence →
6. **Cited QA** — answer + sources

Achieved

End-to-end flow from article text to graph evidence.
Cited QA can show sources and reasoning paths.

Next experiment

Schema–gold audit.
Larger PMCID holdout.
Relation calibration.

Then

Hybrid graph + vector baseline.
Cross-chunk links.
Cost and latency metrics.

Takeaway: the contribution is the data transformation pipeline; the next work should strengthen relation evidence before adding broader retrieval baselines.

Questions?

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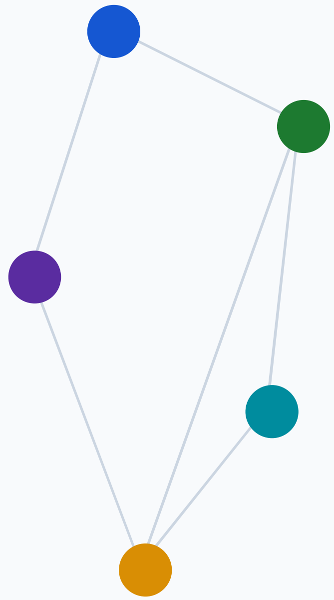
Article → **Graph** → **Evidence** → **Answer**

Traceable graph evidence and cited answers.



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